

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester III Examination, 2023–24

Subject: MATHEMATICS
Paper No. MTB-302: Differential Equations

Time: 3 Hours

Full Marks: 70

Instructions: Answer five questions including Question 1 which is compulsory. The figures in the right-hand margin indicate marks allotted to the respective parts.

1. Answer all parts of this question (Compulsory):

(a) Define the order and degree of a differential equation. [2]

(b) Obtain the relation in a , b , and c (all constants), when $y = xe^{-(\frac{b}{2a})x}$ is a solution of the differential equation:

$$a \frac{d^2y}{dx^2} + b \frac{dy}{dx} + cy = 0$$

[2]

(c) Find an integrating factor for the differential equation:

$$y(axy + e^x)dx - e^x dy = 0$$

[2]

(d) Solve the equation:

$$\frac{dy}{dx} + ay = e^{mx}$$

[2]

where a and m are constants.

(e) Obtain the differential equation of the family of planes represented by:

$$cy = 2 + c^2x$$

[2]

(f) State the standard conditions required for the existence and uniqueness of a solution $y(x)$ for a given regular first-order initial value problem. [2]

2. (a) Formulate and solve the initial value problem for a basic logistics population growth model utilizing the method of separation of variables. [6]

(b) Solve the differential equation:

$$xyp^2 + p(3x^2 - 2y^2) - 6xy = 0 \quad \text{where } p = \frac{dy}{dx}$$

[6]

3. (a) Solve the higher-order linear differential equation with constant coefficients:

$$\frac{d^3y}{dx^3} + 2\frac{d^2y}{dx^2} + \frac{dy}{dx} = e^{2x} + x^2 + x$$

[6]

(b) Solve the following equation:

$$\frac{d^2y}{dx^2} + a^2y = \sec ax$$

[6]

4. (a) If $f(D) \equiv \sum_{r=0}^n a_r D^r$ where a_i ($i = 0, 1, 2, \dots, n$) are constants and $D \equiv \frac{d}{dx}$, then establish the formula for the inverse operator:

$$\frac{1}{f(D)}[\cos ax] = \frac{[f_1(-a^2) \cos ax + af_2(-a^2) \sin ax]}{f_1^2(-a^2) + a^2 f_2^2(-a^2)}$$

where $f(D) \equiv f_1(D^2) + Df_2(D^2)$.

[6]

(b) Using the method of variation of parameters, solve the differential equation:

$$x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} - y = x^2 e^x$$

[6]

External Portals & Academic Resources:

Examination Portal: <https://bhu-exam.vercel.app/> — Student Resource Center: <https://quashan-me.vercel.app/>
Academic Portfolio: <https://me-aryan.vercel.app>